RAJALAKSHMI ENGINEERING COLLEGE
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AD23532 – PRINCIPLES OF DATA SCIENCE
EXPERIMENT 7: MODEL PLANNING AND BUILDING – CLUSTERING

QUESTION:
A retail company wants to group its customers based on Annual Income and Spending Score to develop personalized marketing campaigns. Apply the K-Means Clustering algorithm to identify customer segments, determine the optimal number of clusters using the Elbow Method, visualize the clusters, analyze customer characteristics, and recommend suitable marketing strategies.

AIM:
To perform customer segmentation using the K-Means Clustering algorithm on the Mall Customer Segmentation dataset by exploring the data, standardizing features, determining the optimal number of clusters using the Elbow Method, visualizing customer groups, and analyzing cluster characteristics.

ALGORITHM:
1. Import the required Python libraries: Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.
2. Load the Mall Customer Segmentation dataset and display the first and last five records, dataset shape, data types, missing values, duplicate records, and summary statistics.
3. Select Annual Income (k$) and Spending Score (1–100) as the features for clustering.
4. Standardize the selected features using StandardScaler.
5. Visualize the selected features using scatter plots and pair plots.
6. Apply the Elbow Method to determine the optimal number of clusters by plotting the Within-Cluster Sum of Squares (WCSS).
7. Train the K-Means model using the optimal number of clusters and assign cluster labels to each customer.
8. Visualize the clusters using a scatter plot with different colors and mark the cluster centroids.
9. Calculate the number of customers and the average Annual Income and Spending Score for each cluster.
10. Save the dataset with cluster labels to a new CSV file and suggest suitable marketing strategies for each customer segment.

PROGRAM & OUTPUT:

```python
# Name: Shyam M (241801267)
# Section 1: Import Libraries and Load Dataset

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

df = pd.read_excel("/content/Mall Customers.xlsx")

print("First five records:\n", df.head())
print("Last five records:\n", df.tail())
print("Dataset shape:", df.shape)
print("\nData types:\n", df.dtypes)
print("\nMissing values:\n", df.isnull().sum())
print("Duplicate records:", df.duplicated().sum())
print("\nSummary statistics:\n", df.describe(include="all"))
```

```
First five records:
   CustomerID Gender  Age   Education  Marital Status  Annual Income (k$)  \
0           1      M   19  High School        Married                  15
1           2      M   21     Graduate         Single                  15
2           3      F   20     Graduate        Married                  16
3           4      F   23  High School        Unknown                  16
4           5      F   31   Uneducated        Married                  17

   Spending Score (1-100)
0                      39
1                      81
2                       6
3                      77
4                      40
Last five records:
     CustomerID Gender  Age   Education  Marital Status  Annual Income (k$)  \
195         196      F   35  High School        Married                 120
196         197      F   45  High School        Unknown                 126
197         198      M   32   Uneducated        Married                 126
198         199      M   32  High School        Married                 137
199         200      M   30      College        Married                 137

     Spending Score (1-100)
195                      79
196                      28
197                      74
198                      18
199                      83
Dataset shape: (200, 7)
```

```
Data types:
 CustomerID                int64
Gender                    object
Age                        int64
Education                 object
Marital Status            object
Annual Income (k$)         int64
Spending Score (1-100)     int64
dtype: object

Missing values:
 CustomerID                0
Gender                     0
Age                        0
Education                  0
Marital Status             0
Annual Income (k$)         0
Spending Score (1-100)     0
dtype: int64
Duplicate records: 0

Summary statistics:
        CustomerID Gender         Age Education  Marital Status  \
count   200.000000    200  200.000000       200             200
unique         NaN      2         NaN         7               4
top            NaN      F         NaN   Graduate        Married
freq           NaN    112         NaN        67             112
mean    100.500000    NaN   38.850000       NaN             NaN
std      57.879185    NaN   13.969007       NaN             NaN
min       1.000000    NaN   18.000000       NaN             NaN
25%      50.750000    NaN   28.750000       NaN             NaN
50%     100.500000    NaN   36.000000       NaN             NaN
75%     150.250000    NaN   49.000000       NaN             NaN
max     200.000000    NaN   70.000000       NaN             NaN

        Annual Income (k$)  Spending Score (1-100)
count           200.000000              200.000000
unique                 NaN                     NaN
top                    NaN                     NaN
freq                   NaN                     NaN
mean             60.560000               50.200000
std              26.264721               25.823522
min              15.000000                1.000000
25%              41.500000               34.750000
50%              61.500000               50.000000
75%              78.000000               73.000000
max             137.000000               99.000000
```

```
# Name: Shyam M (241801267)
# Section 2: Select Features

features = ["Annual Income (k$)", "Spending Score (1-100)"]
X = df[features].dropna()
```

```
# Name: Shyam M (241801267)
# Section 3: Standardize Features

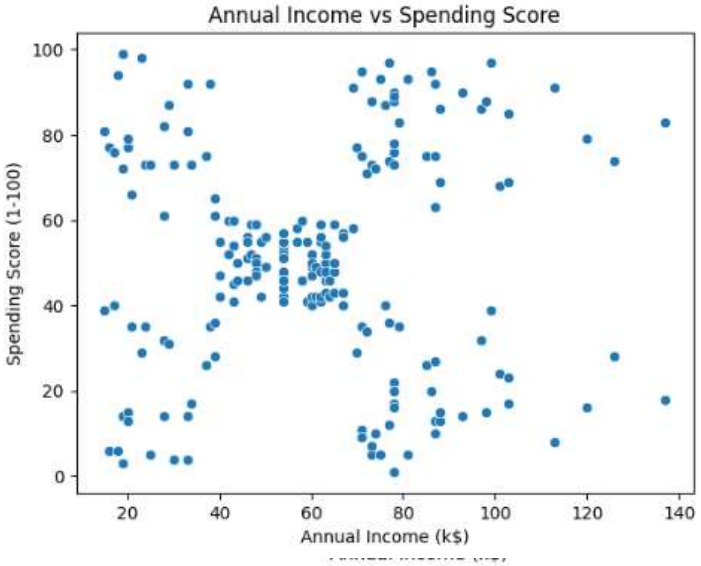
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

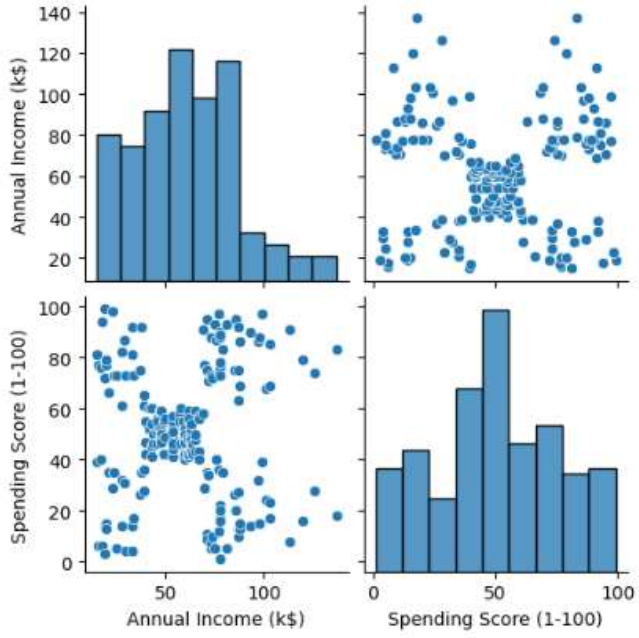
```
# Name: Shyam M (241801267)
# Section 4: Visualize Selected Features

sns.scatterplot(data=df, x=features[0], y=features[1])
plt.title("Annual Income vs Spending Score")
plt.show()

sns.pairplot(df[features].dropna())
plt.show()
```

...



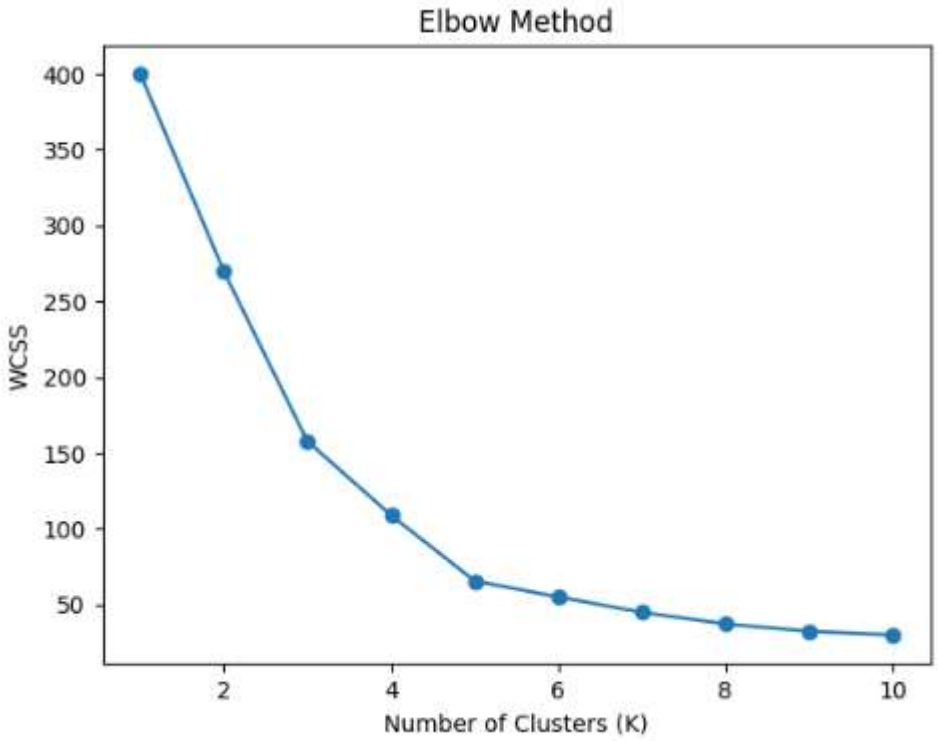


```
# Name: Shyam M (241801267)
# Section 5: Elbow Method

wcss = []

for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

plt.plot(range(1, 11), wcss, marker="o")
plt.title("Elbow Method")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.show()
```




```
# Name: Shyam M (241801267)
# Section 6: Build K-Means Model

optimal_k = 5

kmeans = KMeans(
    n_clusters=optimal_k,
    random_state=42,
    n_init=10
)

df.loc[X.index, "Cluster"] = kmeans.fit_predict(X_scaled)
```

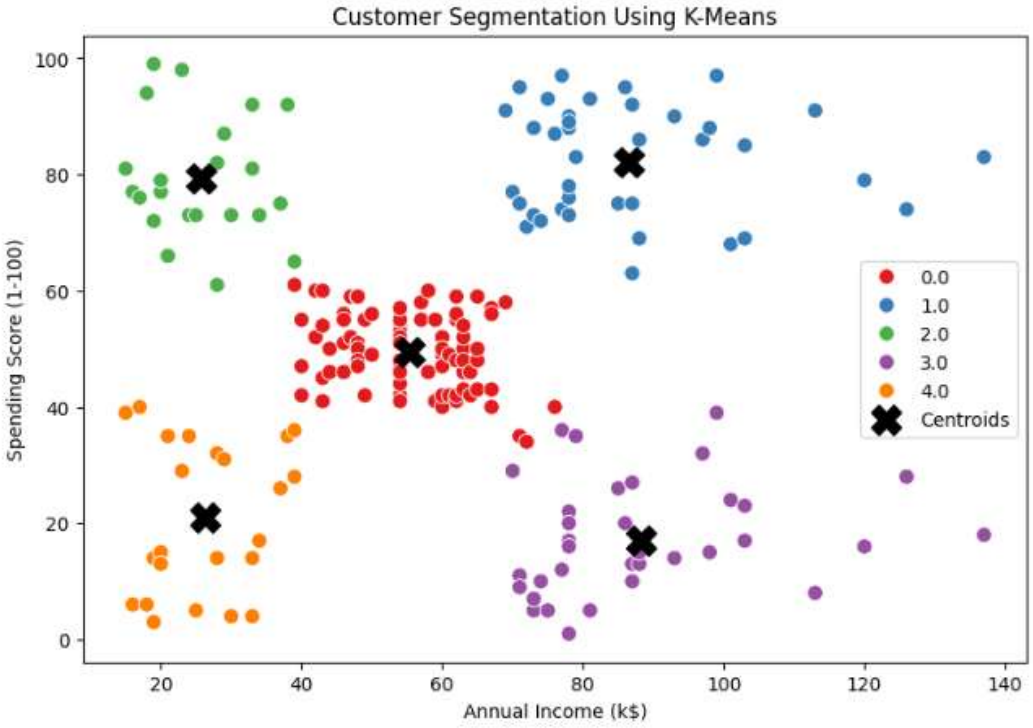
```
# Name: Shyam M (241801267)
# Section 7: Visualize Clusters and Centroids

centers = scaler.inverse_transform(kmeans.cluster_centers_)

plt.figure(figsize=(9, 6))
sns.scatterplot(
    data=df.loc[X.index],
    x=features[0],
    y=features[1],
    hue="Cluster",
    palette="Set1",
    s=70
)

plt.scatter(
    centers[:, 0], centers[:, 1],
    marker="X", s=250, color="black",
    label="Centroids"
)

plt.title("Customer Segmentation Using K-Means")
plt.legend()
plt.show()
```



```
# Name: Shyam M (241801267)
# Section 8: Analyze Clusters

print("Number of Customers in Each Cluster:\n",
      df["Cluster"].value_counts().sort_index())

print("\nAverage Income and Spending Score by Cluster:\n",
      df.groupby("Cluster")[features].mean().round(2))
```

```
Number of Customers in Each Cluster:
 Cluster
0.0    81
1.0    39
2.0    22
3.0    35
4.0    23
Name: count, dtype: int64

Average Income and Spending Score by Cluster:
         Annual Income (k$)  Spending Score (1-100)
Cluster
0.0                   55.30                   49.52
1.0                   86.54                   82.13
2.0                   25.73                   79.36
3.0                   88.20                   17.11
4.0                   26.30                   20.91
```

```
# Name: Shyam M (241801267)
# Section 9: Save Clustered Dataset

df.to_csv("Mall_Customers_Clustered.csv", index=False)
print("Clustered dataset saved successfully.")
```

```
Clustered dataset saved successfully.
```

```
# Name: Shyam M (241801267)
# Section 10: Marketing Recommendations

print("\nMarketing Recommendations:")
print("1. Offer premium rewards to high-income, high-spending customers.")
print("2. Provide targeted promotions to high-income, low-spending customers.")
print("3. Use loyalty offers to retain frequent and high-value customers.")
```

```
Marketing Recommendations:
1. Offer premium rewards to high-income, high-spending customers.
2. Provide targeted promotions to high-income, low-spending customers.
3. Use loyalty offers to retain frequent and high-value customers.
```

+ Code

RESULT:

Thus, customer segmentation was successfully performed on the Mall Customer Segmentation dataset using the K-Means Clustering algorithm. The optimal number of clusters was analyzed using the Elbow Method, and customers were grouped based on Annual Income and Spending Score. The clusters and their centroids were visualized, customer counts and average cluster characteristics were calculated, and suitable marketing strategies were recommended for each customer segment.